

UPI Transaction Insights Dashboard

An end-to-end data analysis project simulating a fintech company's UPI (Unified Payments Interface) transaction data from raw dataset generation through cleaning, SQL analysis, and an interactive Power BI dashboard, culminating in business-facing insights and recommendations.

Built to demonstrate the full data-to-decision workflow expected of a Data/Business Analyst role: generating and cleaning data, writing analytical SQL, building stakeholder-ready visualizations, and translating findings into actionable business recommendations.

Project Overview

- **Dataset:** 9,467 simulated UPI transactions across 2024
- **Tools used:** Python (pandas), SQLite, Power BI, DAX
- **Workflow:** Dataset generation → Data cleaning → SQL analysis → Dashboard → Insight writeup

Folder Structure

Folder	Contents
1_dataset_generation	Python script generating a realistic synthetic UPI transaction dataset
2_data_cleaning	Pandas-based data cleaning: handling nulls, formatting, deduplication
3_sql	7 SQL queries answering key business questions, plus the CSV export script
Power bi	The .pbix Power BI file and page-by-page screenshots
5_insights_summary/	Full written analysis: findings + business recommendations
exports/	CSV outputs of each SQL query, used to power the Power BI dashboard

Dashboard Preview

Page 1 - Executive Overview: Success/failure rates, failure reason breakdown, payment method comparison **Page 2 - Regional & Payment Insights:** City tier performance (volume vs. transaction value) **Page 3 - User Behavior & Segmentation:** Cohort retention trends, RFM user segmentation

(See [4_powerbi_dashboard/screenshots/](#) for page previews)

Key Findings

- **21.35% of transactions fail** - a systemic reliability issue, not specific to any payment method or city tier
- **Network Error + Timeout account for ~38% of failures** - the largest *controllable* share, pointing to infrastructure as the top fix priority
- **Retention drops from 91.56% (January cohort) to 50.33% (June cohort)** - signals a need for stronger new-user onboarding
- **Only 1.2% of users are "High Value"** (RFM segmentation), yet they likely drive disproportionate transaction value - a strong case for a targeted loyalty program
- **Tier 3 cities have the highest average transaction value** despite the lowest volume an underleveraged growth opportunity

Full analysis with business recommendations:

[5_insights_summary/insights_summary.md](#)

Tech Stack

- **Python** - dataset generation, data cleaning (pandas)
 - **SQLite** - analytical querying
 - **Power BI** - dashboarding and DAX measures
 - **Git/GitHub** - version control and portfolio hosting
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